CS527: Software Testing and Analysis
Course Projects

Aug 27, 2026

In CS527: Software Testing and Analysis (Fall 2026), the course project mainly focuses on building **LLM-based** software quality assurance (QA) techniques.

1. QA task candidate

You may select to build one of the following QA techniques, including but not limited to: (1) program analysis techniques that reasons certain properties (e.g., functional, safety, non-functional) of the program; (2) test generation techniques (e.g., unit test or fuzzing) that exploit or detect vulnerabilities/bugs in software*;* (3) debugging techniques (e.g., fault localization, program repair, issue resolution) that mitigate or fix the bugs in software; (4) security analysis techniques (e.g., vulnerability detection).

Note: any previously-unknown bugs or zero day vulnerabilities detected by your technique can gain additional credits

2. Technique design

No matter which QA task you plan to tackle, please ensure that the technique is based on generative AI. For example, it could be a hybrid approach that combines LLM and traditional analysis approach; it could be a LLM-driven workflow; it could be an agentic approach.

3. Software Scope

The technique could be designed for domain-specific software (e.g., Linux kernel, compiler, AI compiler, DL libraries, AI agents) or could be a general approach that tackles a wide range of software systems.

4. Grading

The course project takes 50% of the entire course score. In particular, the course project is graded with the following items, and we evaluate each item based on both the involved effort/workload and final results you get.

- Proposal report/presentation (10%): 1-page proposal and 5 min presentation. The proposal should include: (1) target problem and motivation, (2) plan for approach design, (3) plan for evaluation design, (4) expected outcome.

- Midterm report/presentation (10%): 3-page PDF report and 10 min presentation. The midterm report should include: (1) your progress, (2) faced challenges and solutions, (3) changes made since proposal, (4) plan for final report.

- Final report/presentation (30%): 5-page PDF report and 10 min presentation. The final report should include: (1) introduction (target problem and motivation), (2) related work, (3) approach or study methodology, (4) evaluation (Settings and Results), (5) discussion (limitations and future extension).

You are encouraged to complete the course project in groups of 3–5.